

PUSHPAK V1

Tilt-Rotor VTOL UAV Demonstrator

Mission: demonstrate electrical-power reduction in forward flight through wing-borne lift vs pure rotor-borne flight.

4 ROTORS

FRONT + REAR TILT PAIRS

4.5 kg

DESIGN-POINT MTOW

1.0 kg

PAYLOAD

PIXHAWK

ARDUPILOT + ESP32

PROJECT TEAM

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Project Overview

RC tilt-rotor VTOL demonstrator · four rotors · fixed wing

Prove that wing-borne lift reduces electrical power in forward flight relative to pure rotor-borne flight at the same MTOW.

AIRCRAFT

RC tilt-rotor VTOL
4 rotors + fixed wing
Front & rear tilt pairs

DESIGN POINT

MTOW 4.5 kg
Target 4.0–4.5 kg
Payload 1.0 kg

PROPULSION

4 × MY4215 400KV
16-inch class props
6S Li-ion ~20 Ah

AVIONICS

Pixhawk + ArduPilot
ESP32 companion
Pitot · GPS · mag encoders

TILT

2 × STS3215-C018
One servo per axis
Max tilt 75°

GEOMETRY

Wing span 1300 mm
Fuselage 1200–1300 mm
Rotor span 1700–1900 mm

Problem Statement

CONSTRAINT

Rotor-borne flight carries the full weight through the propellers.

- ▶ Induced power scales with disk loading, not with airspeed.
- ▶ A 4.5 kg quadrotor still burns hundreds of watts just to stay aloft.
- ▶ Edgewise forward flight adds profile and parasite penalties.
- ▶ Endurance and range collapse before a useful payload mission is complete.
- ▶ Student-class VTOL demonstrators rarely isolate the wing-lift benefit with flight data.

RESPONSE

Add a wing. Offload lift. Measure the electrical power that disappears.

- ▶ Wing target: 70% of weight at 20 m/s cruise.
- ▶ Rotors unload, then tilt to 75° to provide thrust.
- ▶ Same airframe flown wing-on vs wing-off (or rotor-borne equivalent).
- ▶ Success = a documented reduction in electrical power, not a paper polar.
- ▶ Demonstrator mass, cost and complexity stay inside a university build.

Why VTOL + Fixed Wing

Attribute	Multicopter	Fixed wing	Tilt-rotor VTOL
Runway	None	Required	None
Hover	Native	No	Native (4 rotors)
Cruise lift	Rotor-borne	Wing-borne	Wing + residual rotor
Cruise power	High	Low	Target: reduced
Propulsion sets	1	1	1 (reused)
Transition	N/A	N/A	Scheduled tilt + throttle
Instrumentation	Easy	Easy	Tilt angle + airspeed + power
Fit to mission	No (no wing benefit)	No (no VTOL)	Selected

Selection logic · one motor set · two tilt axes · measurable wing-offload · ArduPilot VTOL heritage

Mission Requirements

MR-1

PRIMARY

Quantify electrical-power reduction in wing-borne forward flight versus rotor-borne flight at the same mass.

MR-2

VTOL

Vertical take-off and landing from an unprepared 10 × 10 m pad, no catapult, no net.

MR-3

PAYLOAD

Carry 1.0 kg payload at design-point MTOW 4.5 kg (empty aircraft ≤ 3.5 kg).

MR-4

CRUISE

Sustain 20 m/s with the wing producing ~70% of lift ($CL \approx 0.395$ on 0.32 m^2).

MR-5

TRANSITION

Stable hover → 75° tilt transition using Pixhawk / ArduPilot with logged states.

MR-6

DATA

Log airspeed, tilt, RPM, bus voltage/current, actuator current, and inertial rates.

Design Objectives

01

Power experiment

Wing-on vs wing-off (or rotor-borne) electrical-power comparison at matched mass and speed.

02

Thrust margin

$\geq 2.0\times$ hover thrust (target 9 kgf static on 4.5 kg).

03

Wing-offload

NACA 4412, 0.32 m^2 , AR 5.3, CL 0.395 at 20 m/s.

04

Tilt authority

Front and rear axes independent, $0\text{--}75^\circ$, magnetic angle closed-loop.

05

Open-source FC

ArduPilot VTOL stack on Pixhawk; ESP32 for high-rate tilt / power I/O.

06

Buildable

University workshop: 3D-printed mounts, composite/foam wing, COTS propulsion.

07

Traceable mass

Empty $\leq 3.5\text{ kg}$; live mass budget from CAD + weigh-ins.

08

Safe envelope

Mechanical tilt stops, current limits, geofence, RTL, dual-axis jam detect.

System Requirements

ID	Requirement	Target	Verification
SR-P1	Hover thrust per motor	≥ 1.125 kgf (design 2.25)	Static thrust stand
SR-P2	Total static thrust	≈ 9 kgf	4-motor stand, 6S
SR-P3	Propulsion mass	Motors 0.80 kg; ESC 0.25 kg	Scale + CAD
SR-A1	Wing area / AR / airfoil	$0.32\text{ m}^2 \cdot 5.3 \cdot \text{NACA } 4412$	CAD + template
SR-A2	Cruise CL at 20 m/s	≈ 0.395 (70% weight)	CFD + pitot + load
SR-A3	Rotor / wing clearance	50-75 mm min	CAD + ground spin
SR-C1	Tilt range / axes	$0\text{-}75^\circ \cdot \text{front} + \text{rear}$	Encoder log
SR-C2	Flight stack	Pixhawk + ArduPilot + ESP32	HIL + taxi
SR-T1	Power comparison	Wing-on vs wing-off	Flight / trolley test
SR-S1	Empty mass	≤ 3.5 kg	Weigh-in

Aircraft Configuration Selection

A TAILSITTER

Single attitude change. Poor hover precision. Hard to instrument a clean wing-off comparison.

B TILT-WING

Wing + rotors tilt together. High hinge moment, complex structure, poor student-build risk.

C DUAL-SYSTEM

Lift rotors + cruise prop. Two motor sets. Mass and power electronics kill the 3.5 kg empty target.

SELECTED

D TILT-ROTOR

Four rotors, two tilt shafts. Hover as a quad. Cruise on the wing. One propulsion set. **SELECTED.**

FROZEN CONFIGURATION

4 rotors · front pair tilts together · rear pair tilts together · max tilt 75° · high wing + 2 winglets · Pixhawk / ArduPilot

Overall Aircraft Architecture



AIRFRAME

Fuselage 1.2–1.3 m
Wing 1.30 m · 0.32 m²
Winglets × 2 · skids

PROPULSION

4 × MY4215 400KV
16-in class · 6S
Target 9 kgf static

TILT

2 shafts · 2 servos
8 bearings · stops
Mag angle sensors

ENERGY

6S Li-ion ~20 Ah
1.0–1.5 kg pack
Current / voltage log

GNC

Pixhawk + ArduPilot
ESP32 companion
Pitot · GPS · IMU

PAYLOAD

1.00 kg dummy
CG hard point
Not a sensor suite

FRONT AXIS

Tilt together
Left + right motors

REAR AXIS

Tilt together
Left + right motors

Propulsion System

4 ×

HGLRC MY4215

400 KV · 200 g each

800 g

TOTAL MOTOR MASS

18% of 4.5 kg MTOW

16 in

PROPELLER CLASS

D = 406 mm

6S

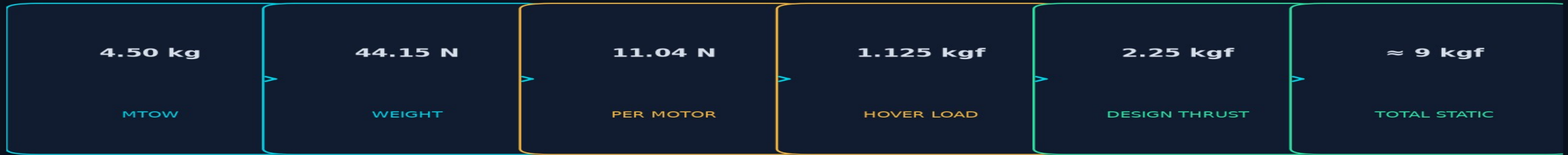
LI-ION PACK

Initial concept ~20 Ah

Item	Spec	Qty	Role
Motor	HGLRC MY4215 400KV	4	Hover + cruise thrust
Motor mass	200 g (800 g set)	4	Mass budget line
Propeller	16-inch class / 406 mm	4	Disk area for hover
ESC	From 0.25 kg allocation	4	6S capable, logged current
Battery	6S Li-ion ~20 Ah	1	Energy + CG
Bus	6S power distribution	1	Voltage / current sense
Mounts	Tilt-shaft motor mounts	4	Interface to tilt axis

Hover Thrust Analysis

MTOW 4.5 kg · g = 9.81 m/s² · four equal motors



W = MG

$$4.50 \times 9.81 = 44.15 \text{ N}$$

T_HOVER / MOTOR

$$44.15 / 4 = 11.04 \text{ N}$$

KGF EQUIVALENT

$$11.04 / 9.81 = 1.125 \text{ kgf}$$

DESIGN T / MOTOR

$$2.25 \text{ kgf} \quad (\approx 22.1 \text{ N})$$

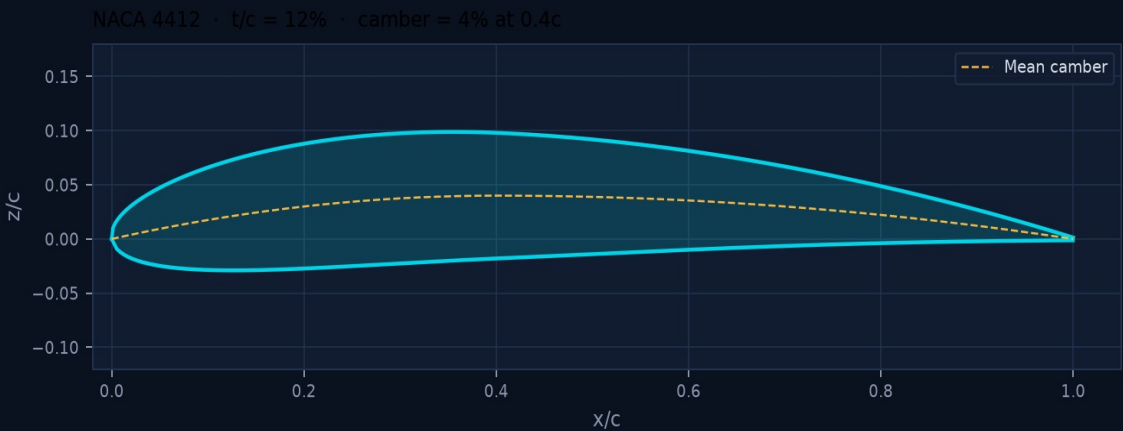
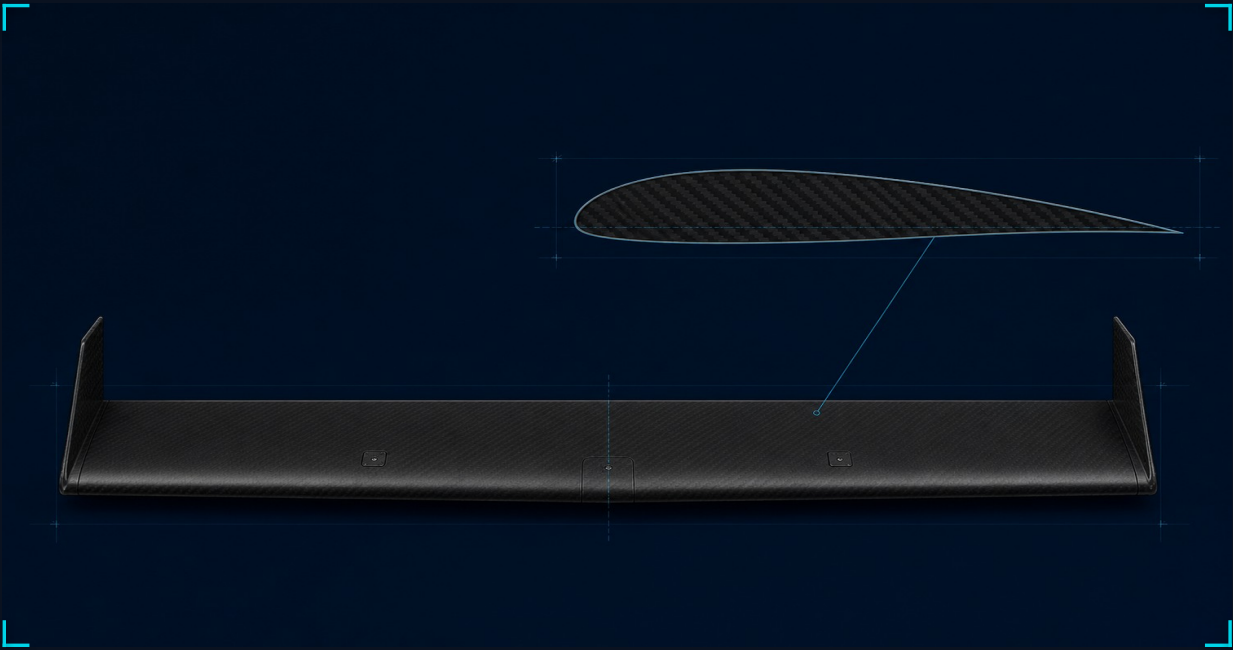
TOTAL STATIC T

$$\approx 9.0 \text{ kgf} \quad (\approx 88 \text{ N})$$

THRUST / WEIGHT

$$9.0 / 4.5 \approx 2.0$$

Wing Design Methodology



SPAN	AREA	CHORD	AR	FOIL	TIPS
1300 mm	0.32 m ²	246 mm	5.3	4412	2 winglets

Lift Calculations

CRUISE LIFT EQUATION

$$L = \frac{1}{2} \rho V^2 S C_L \quad \rightarrow \quad C_L = L / (q S)$$

$$\rho = 1.225 \text{ kg/m}^3 \cdot V = 20 \text{ m/s} \cdot q = 245 \text{ Pa} \cdot S = 0.32 \text{ m}^2 \cdot L_{\text{target}} = 0.70 \times 44.15 = 30.90 \text{ N}$$

20 m/s

CRUISE SPEED

70%

WING LIFT SHARE

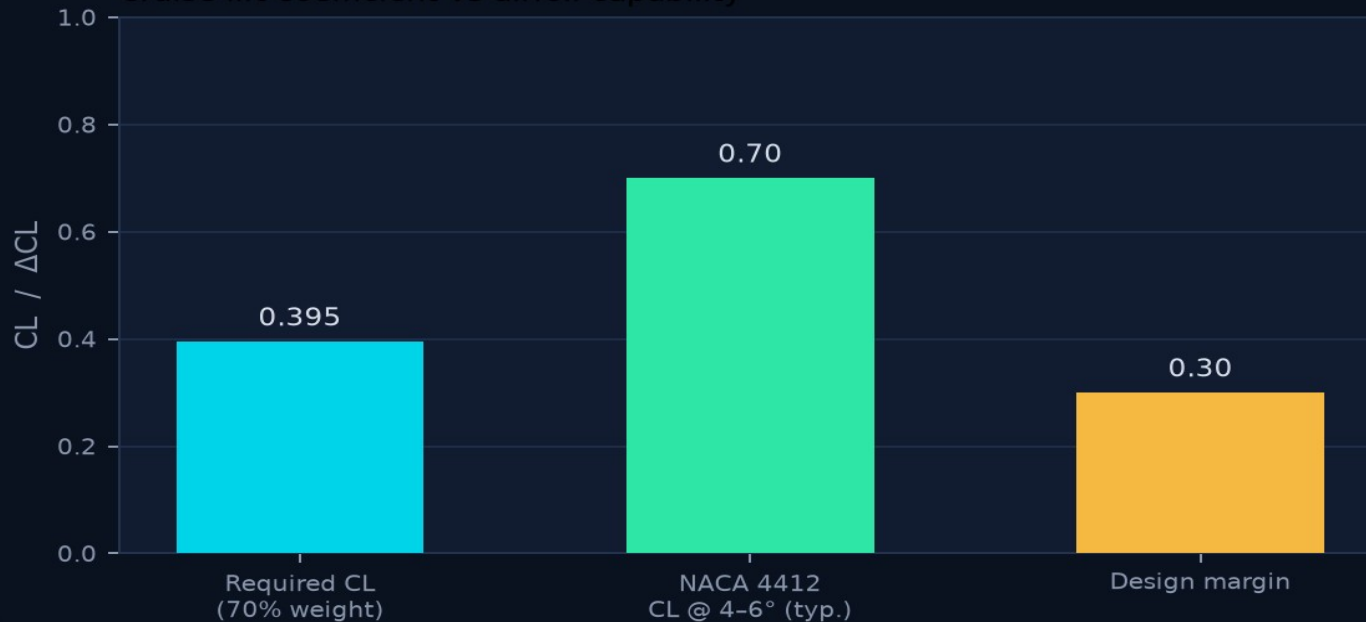
0.395

REQUIRED CL

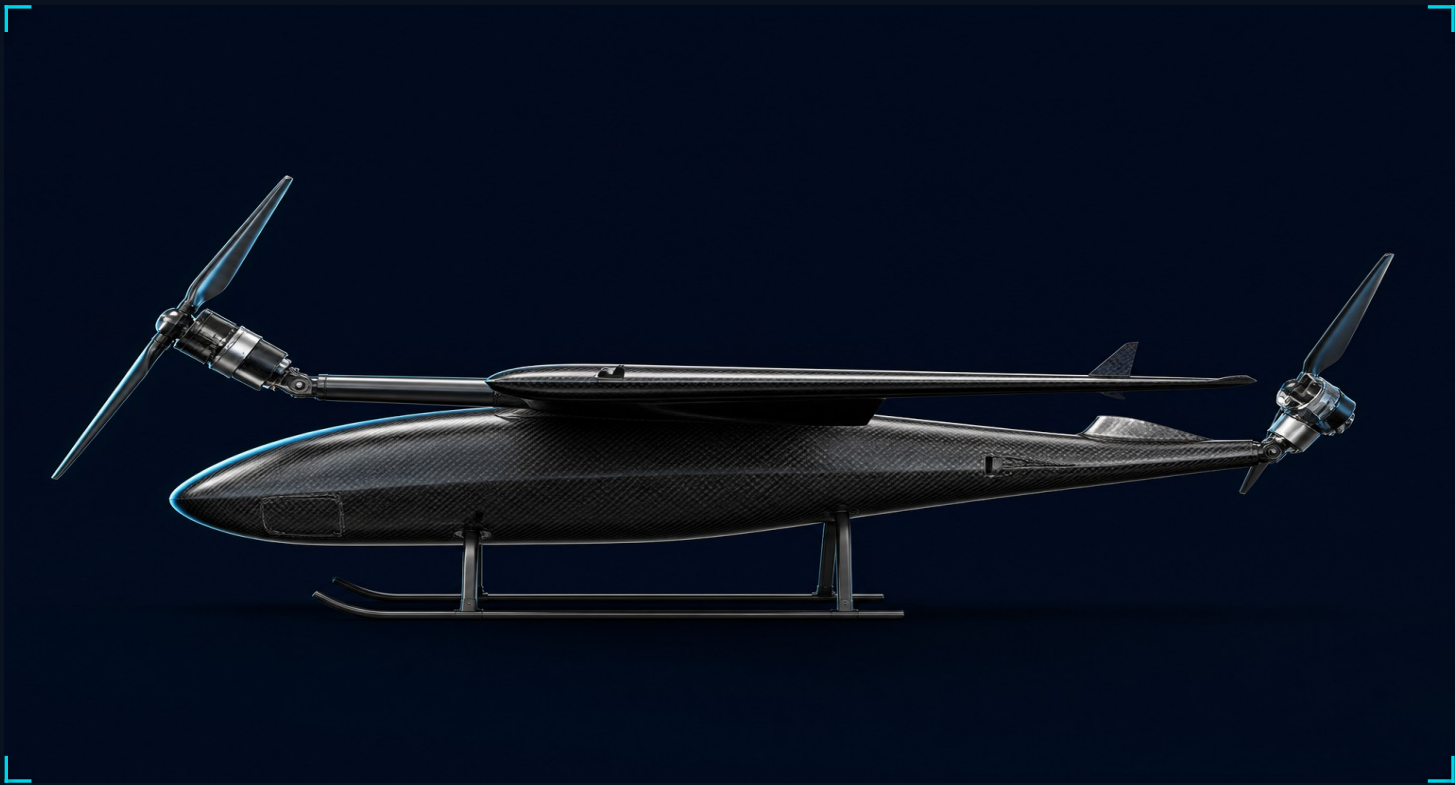
30.90 N

WING LIFT

Cruise lift coefficient vs airfoil capability



Aircraft Geometry

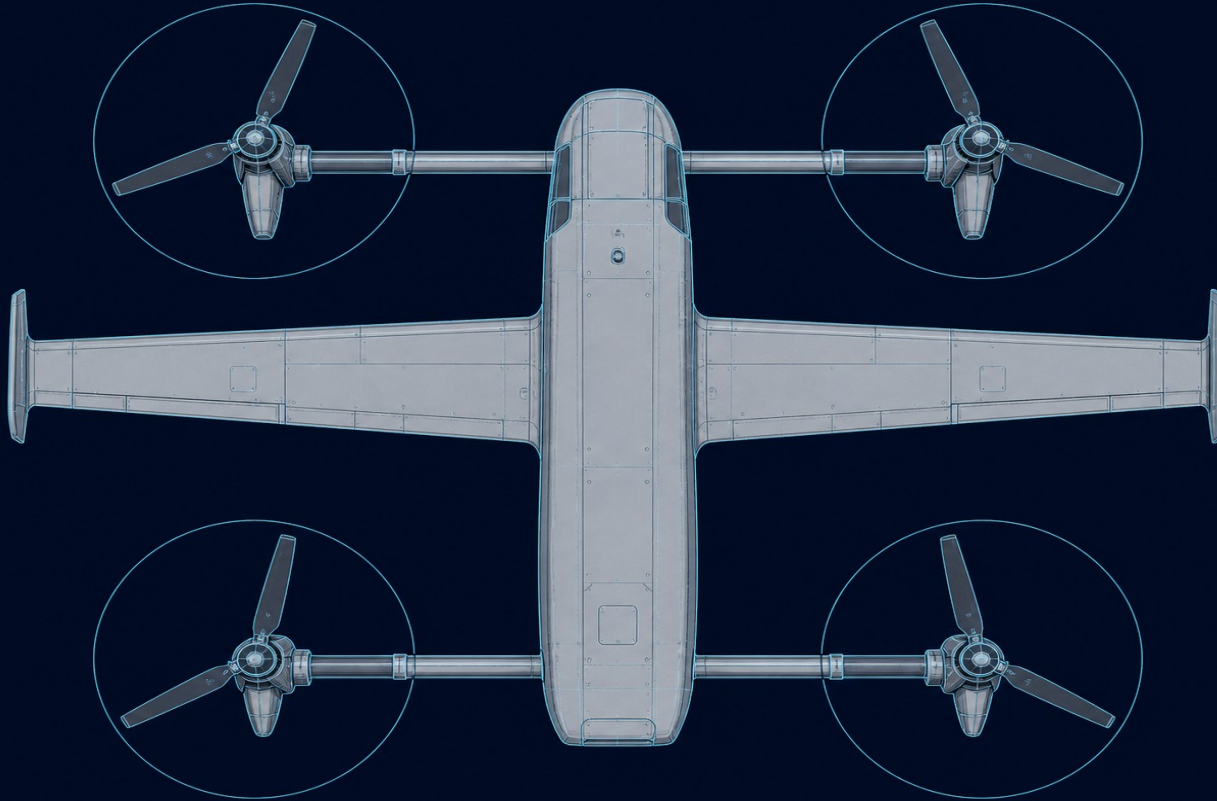


Station	Dimension
Fuselage length	1200–1300 mm
Wing span	1300 mm
Rotor span (tip-to-tip)	1700–1900 mm
Propeller diameter	406 mm
Mean chord	246 mm
Wing area	0.32 m ²
Max rotor tilt	75°
Min prop clearance	50–75 mm

LAYOUT INTENT

High wing keeps the prop disks clear of the fuselage and simplifies landing-gear length. Front tilt shaft sits ahead of the leading edge; rear shaft sits aft of the trailing edge. CG is between the two shafts so hover moments stay small.

Rotor Placement Layout



FRONT LEFT

On front tilt shaft · tilts with FR

FRONT RIGHT

On front tilt shaft · 1 servo

REAR LEFT

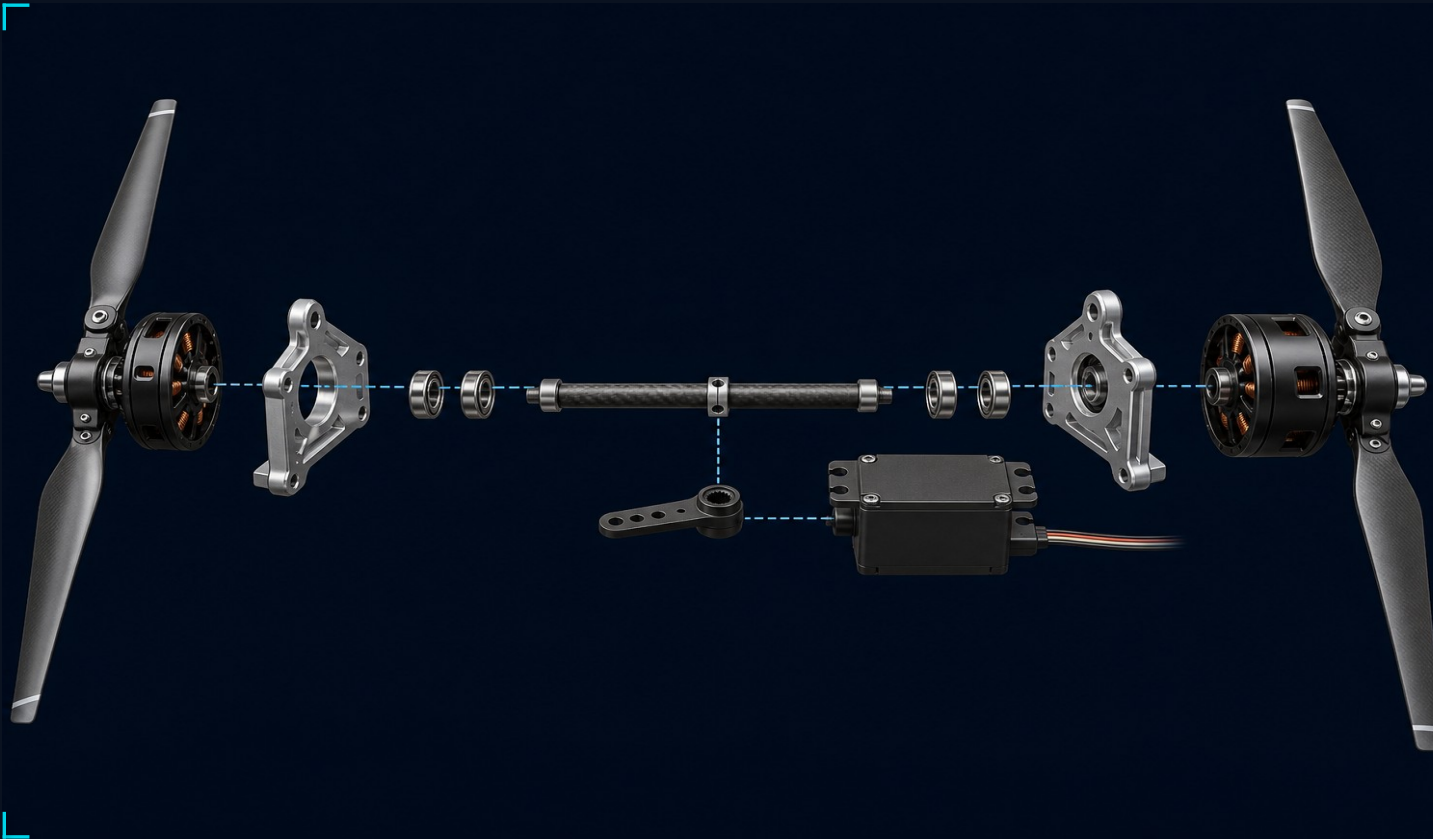
On rear tilt shaft · tilts with RR

REAR RIGHT

On rear tilt shaft · 1 servo

Minimum propeller clearance 50–75 mm · rotor span 1700–1900 mm · propeller D 406 mm · wing span 1300 mm (between the two shafts)

Tilt Mechanism Architecture



PER AXIS

1 shaft · 2 motor mounts · 4 bearings · 1 servo · linkage · stops · 1 angle sensor

ACTUATOR

STS3215-C018 serial bus servo · one per axis · not per motor

STOPS

Hard mechanical limits at hover (0°) and cruise (75°)
· independent of software

SENSOR

Absolute magnetic angle · logged to Pixhawk / ESP32 · jam detection

MOTOR

MOUNT

SHAFT

BEARINGS

LINKAGE

SERVO

Tilt Mechanism Working Principle



01 HOVER

Tilt 0° · T vertical
Wing lift ≈ 0
Quad mixer active

02 TRANSITION

Tilt scheduled vs airspeed
Wing loading rising
Collective reducing

03 CRUISE

Tilt 75° · T mostly axial
Wing $\sim 70\%$ of weight
Power experiment window

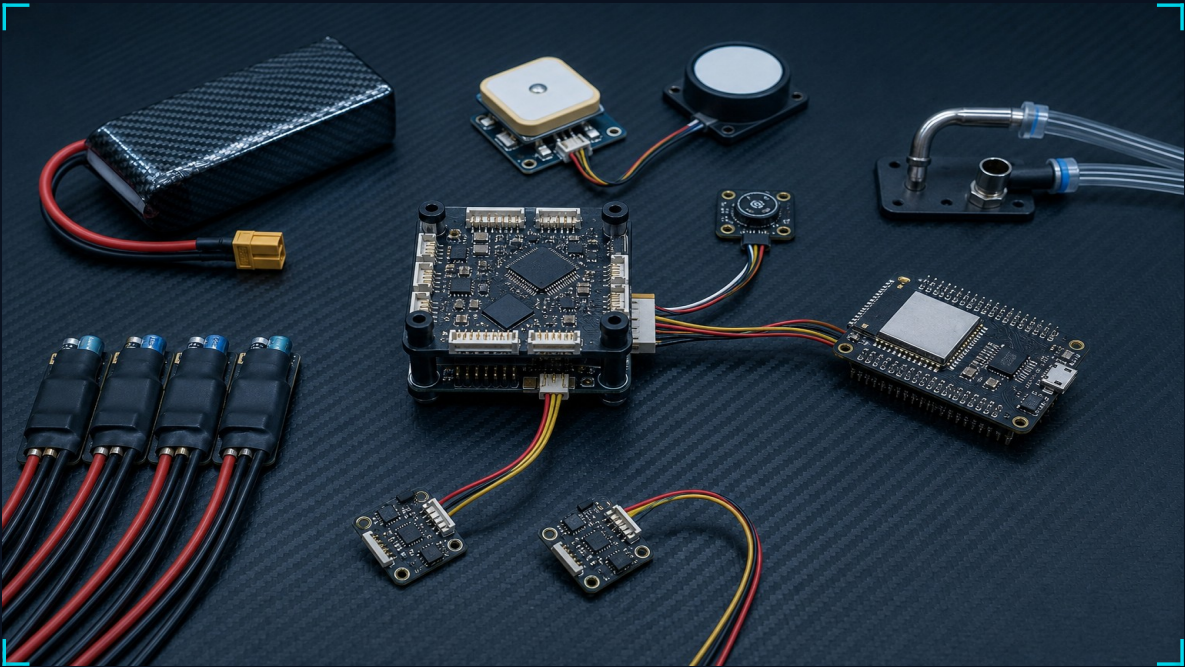
LOAD PATH

COMMAND PATH

MOTOR → MOUNT → TILT SHAFT → BEARINGS → AIRFRAME

FC → SERVO → LINKAGE → SHAFT (servo never carries rotor loads)

Electronics Architecture



SENSE

Pitot · GPS · compass · mag angle ×2 · IMU · V/I bus · actuator current

DECIDE

Pixhawk / ArduPilot flight laws · ESP32 companion I/O and experiment log

ACT

4 × ESC / motor · front tilt servo · rear tilt servo

Node	Function
Pixhawk	Attitude, VTOL mixer, failsafes, logging
ArduPilot	QHOVER / QLOITER / FBWA / AUTO
ESP32	Tilt telemetry, power DAQ, experiment flags
Pitot	Calibrated airspeed for tilt schedule
GPS + compass	Position, heading, RTL
Mag angle	Absolute tilt on each shaft

Pixhawk-ESP32 Integration

PIXHAWK / ARDUPILOT

- ▶ Owns motor mixer and VTOL modes.
- ▶ Runs attitude / position loops.
- ▶ Issues front and rear tilt PWM / bus commands.
- ▶ Hosts failsafes: batt, GCS, geofence, RTL.
- ▶ Records onboard .bin log (IMU, GPS, RC).
- ▶ Reads pitot as airspeed sensor.
- ▶ If companion dies → freeze last tilt or revert to hover.



ESP32 COMPANION

- ▶ Samples magnetic tilt encoders at high rate.
- ▶ Samples battery voltage, current, actuator current.
- ▶ Stamps experiment events (wing-on / wing-off run IDs).
- ▶ Forwards compact telemetry to GCS / SD.
- ▶ Does not command motor throttle.
- ▶ Watchdog heartbeats to Pixhawk.
- ▶ Can inhibit further tilt if encoder disagreement.

Control System Architecture

INPUTS

Pilot command

Airspeed (pitot)

Tilt angle ×2

Vertical velocity

Altitude

RPM ×4

Battery voltage

Battery current

Actuator current

Roll rate

Yaw rate

Vert. acceleration

FLIGHT MODE LOGIC

- ▶ Hover mixer (quad X / H).
- ▶ Tilt schedule vs calibrated airspeed.
- ▶ Front / rear tilt may use a slight split to manage pitch.
- ▶ Collective reduction as wing lift rises.
- ▶ Power experiment flag (wing-on / wing-off).
- ▶ Failsafe overlay (always on).
- ▶ ArduPilot VTOL state machine + ESP32 inhibit.

OUTPUTS

Front tilt command

Rear tilt command

Motor throttle ×4

Flight mode logic

Sensor Suite

PITOT

Calibrated airspeed

Transition gate · CL reconstruction

GPS

Position / groundspeed

Geofence · RTL · track

COMPASS

Heading

Yaw reference in hover & cruise

MAG ANGLE ×2

Shaft tilt

Closed-loop tilt · jam detect

IMU (FC)

Rates + accel

Att. loop · n_z during transition

BARO (FC)

Altitude / V_z

Climb hold in QHOVER / QLOITER

V / I BUS

Electrical power

Primary experiment observable

I_ACTUATOR

Servo / ESC current

Jam, stall, ESC health

Flight Modes

01

QSTABILIZE

Manual throttle, attitude stabilize. Tilt locked at 0°. First tether hover.

02

QHOVER

Alt-hold hover. Used for CG check, tilt-servo health, and pad ops.

03

QLOITER

Position hold. Precision take-off and landing once GPS is trusted.

04

TRANSITION

Airspeed-scheduled tilt 0→75°. Collective reduced as wing unloads rotors.

05

FBWA / CRUISE

Wing-borne science window. Power logging at 20 m/s. Tilt held at 75°.

06

RTL / QRTL

Failsafe home. Tilt back toward hover, then vertical descent on pad.

Transition Strategy

Pad hover

Nose into wind

Build V

Tilt ramp

Wing hold

Cruise log

T+0

QHOVER 1.5–2 m AGL. Confirm mag-angle 0° , bus healthy, pitot alive.

T+1

Command a small positive pitch / forward cyclic. Groundspeed rising, $V_z \approx 0$.

T+2

At V_1 (to be set in taxi tests, expected 8–12 m/s) start tilt ramp, limited $^\circ/\text{s}$.

T+3

As CL builds, lower collective so that altitude does not balloon.

T+4

Hold 75° once $V \approx 20$ m/s and n_z is quiet. Open the power-log window.

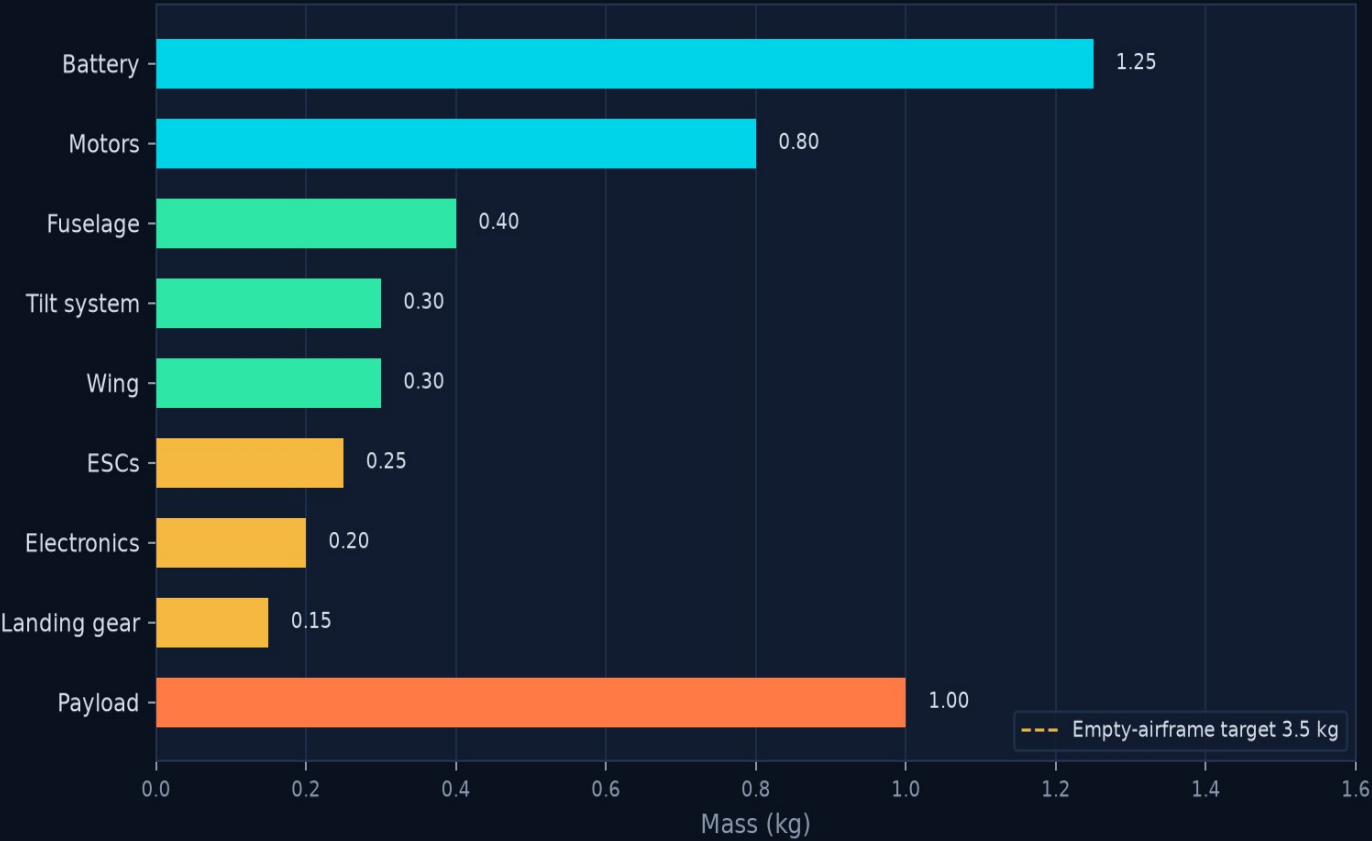
T+5

Abort line: airspeed decay, tilt jam, or $V_z > \text{band}$ $\rightarrow$ reverse tilt, QHOVER.

Safety Features

Layer	Feature	Intent
Mechanical	Hard stops at 0° and 75°	Blade / wing collision impossible via software fault
Mechanical	Shaft bearings carry rotor loads	Servo is a commander, not a structure
Sense	Dual mag-angle disagreement	Inhibit further tilt; hold last safe angle
Sense	Pitot fail detect	Inhibit transition; remain in VTOL modes
Electrical	Bus undervoltage	QRTL / land
Electrical	Actuator overcurrent	Freeze tilt; alert GCS
FC	Lost GCS / geofence	QRTL
FC	Companion watchdog	ESP32 death does not kill motors
Ops	Tethered first hover	Contain first-flight energy
Ops	Prop clearance 50–75 mm	Flex and downwash margin

Mass Budget



Group	Mass (kg)
Payload	1.00
Motors	0.80
ESCs	0.25
Battery	1.0-1.5
Tilt system	0.30
Wing	0.30
Fuselage	0.40
Electronics	0.20
Landing gear	0.15
Estimated MTOW	4.4-4.9
Empty target	≤ 3.50
Design point	4.50

Component Selection Matrix

Propulsion · tilt · flight control · electronics · battery · airframe

PROPULSION

Item	Spec	Qty
Motor	HGLRC MY4215 400KV	4
Propeller	16-inch class / 406 mm	4
ESC	6S, 0.25 kg set	4

TILT MECHANISM

Item	Spec	Qty
Servo	STS3215-C018	2
Tilt shaft + mounts	Front / rear axis	2+4
Bearings / stops	4 per axis + hard stops	8+

FLIGHT CONTROL

Item	Spec	Qty
FC	Pixhawk	1
Stack	ArduPilot VTOL	1
Companion	ESP32	1

ELECTRONICS

Item	Spec	Qty
Airspeed	Pitot probe + sensor	1
Nav	GPS + compass	1+1
Tilt sense	Abs. magnetic angle	2

BATTERY SYSTEM

Item	Spec	Qty
Pack	6S Li-ion ~20 Ah	1
Sense	Bus V and I	1
Mass alloc.	1.0–1.5 kg	—

AIRFRAME

Item	Spec	Qty
Wing	1300 mm NACA 4412	1
Winglets	Tip treatments	2
Fuselage / gear	1200–1300 mm / skids	1

CAD Development Plan



CFD Validation Plan

C1 WING POLAR

Isolated wing + winglets. α sweep at $Re \approx 3.3 \times 10^5$ ($c=0.246$ m, 20 m/s). Extract CL , CD , CM . Confirm $CL=0.395$ is a comfortable α .

C2 HOVER WASH

Four 16-inch disks as actuator disks or MRF, wing at 0° tilt. Map download penalty and elevator blanking.

C3 CRUISE

Full airframe, rotors at 75° , $V=20$ m/s. Residual rotor lift vs wing lift split. Nacelle / boom drag.

C4 TRANSITION

One or two snapshots at $30-45^\circ$ tilt. Identify pitch-up/pitch-down and a safe ramp rate.

SUCCESS METRICS

- ▶ CL_{req} 0.395 inside linear range.
- ▶ Wing stall $\alpha > \text{cruise } \alpha + 6^\circ$.
- ▶ Download increment quantified.
- ▶ No hidden pitch-up at $30-45^\circ$.
- ▶ Drag polar for power estimate.
- ▶ Mesh / residual report in annex.

Structural Analysis Plan

ID	Load case	Factor / condition	Pass criterion
LC-1	Hover thrust on mounts	2.25 kgf × 2.0 per motor	No yield; bolt bearing OK
LC-2	Tilt-shaft torsion / bending	Max motor torque + gyro	Deflection does not eat clearance
LC-3	Wing +2.5 / -1.0 g	Cruise 20 m/s, 70% lift	Spar cap / foam shear OK
LC-4	Landing 3.5 g spike	Skids, fuselage keel	No FC tray crush
LC-5	Servo linkage stall	STS3215 stall torque	Horn / pin shear OK; stop takes load
LC-6	Prop burst / FOD	Containment awareness	Layout keeps blades off boom/wing
LC-7	Modal (shaft + wing)	Hover RPM range	No 1/rev coincidence
LC-8	CG / hard-point	1 kg payload 10 g	Insert pull-out margin

Prototype Manufacturing Plan

01

PROPULSION

Source MY4215, 16-in props, 6S ESCs. Thrust-stand characterisation before airframe lock.

02

TILT AXIS

Shaft, printed/CNC mounts, bearings, stops, STS3215, mag targets. Bench 0–75° cycle test.

03

WING

Hot-wire or CNC foam core, spars, NACA 4412 templates, bonded winglets. Mass $\leq$ 0.30 kg.

04

FUSELAGE

Pod for battery, Pixhawk, payload. Skids. Harness channels. Access panels.

05

INTEGRATION

Harness, pitot boom, GPS, encoder routing, CG weigh, control-surface or trim if any.

06

QA

Prop clearance 50–75 mm, tilt-stop inspection, insulation, photo log, mass report.

Ground Testing Plan

ID	Test	Setup	Pass
GT-1	Motor / prop map	Single-motor stand, 6S	≥ 2.25 kgf / motor; temps OK
GT-2	4-motor hover T	Frame locked, pads	≥ 9 kgf total; vibration log
GT-3	Tilt kinematics	Axis jig, encoder	0-75°, no bind, stop contact
GT-4	Tilt endurance	500 cycles loaded	No play, no mag-angle drift
GT-5	Servo stall / current	Locked horn	Current trip works
GT-6	Pitot calibration	Wind / car boom	IAS vs reference
GT-7	Power DAQ	Known shunt load	W within 5%
GT-8	EMI / compass	Motors spinning	Heading stable
GT-9	Failsafe dry-run	FC on bench	RTL, batt, GCS loss
GT-10	Tether hover	Outdoor, short tether	QHOVER 30 s, no oscillation

Flight Testing Plan

Sortie	Objective	Mode	Abort
FT-1	Hover 30-60 s, CG, vibration	QHOVER	Oscillation / mag-angle noise
FT-2	Pad ops, QLOITER box	QLOITER	GPS / compass disagree
FT-3	Low-speed translation, tilt 0°	QHOVER	Pilot PIO
FT-4	Partial tilt to 20-30°	Transition	Pitch upset, Vz band
FT-5	Full 75° to 20 m/s, short	Trans. + FBWA	Airspeed decay
FT-6	Wing-on power point 20 m/s	CRUISE	Current / temp
FT-7	Wing-off / rotor-borne compare	VTOL fwd	Power > limit
FT-8	Repeatability ×3	FT-6 profile	Inconsistent IAS
FT-9	QRTL from cruise	QRTL	Tilt reverse stall
FT-10	Demo for faculty	Rehearsed profile	Any red abort

Expected Results

2.0×

HOVER T/W

9 kgf static on 4.5 kg

0.395

CRUISE CL

70% weight at 20 m/s

~40–55%

POWER CUT (EST.)

Wing-on vs wing-off fwd

≤ 3.5 kg

EMPTY MASS

Hard engineering target

75°

TILT STOP

Encoder-confirmed

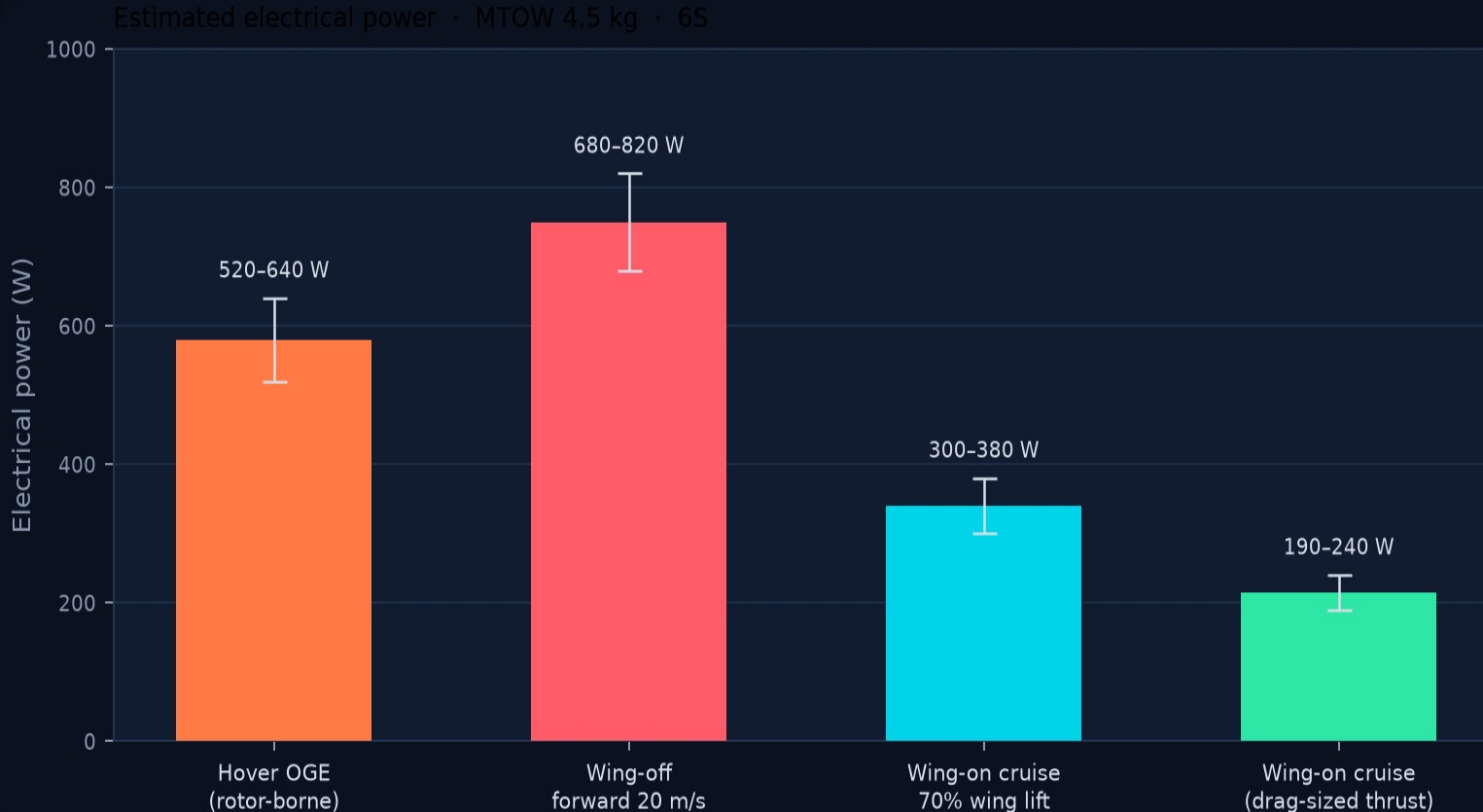
1.0 kg

PAYLOAD

Carried at design point

Wing-On vs Wing-Off Power Comparison

Estimates only · MTOW 4.5 kg · 6S · 16-inch class · $V = 20$ m/s



WHAT WE WILL FLY

- ▶ Same mass, same pack, same day.
- ▶ Stabilised 20 m/s run, level.
- ▶ Log $W = V \times I$, IAS, tilt, RPM.
- ▶ Three repeats each config.
- ▶ Report mean, min, max — not a single heroic pass.
- ▶ If removable wing is unsafe, lock tilt at 0° for the rotor-borne leg.

Success criterion · statistically lower electrical power in the wing-on condition at matched MTOW and airspeed · uncertainty reported

Risks and Mitigation

ID	Risk	L	C	Mitigation
R1	MTOW > 4.9 kg (battery / prints)	M	H	Weekly CAD mass; foam wing; pack downsize
R2	Static thrust < 2.25 kgf / motor	M	H	Stand map before CAD freeze; prop pitch options
R3	Transition pitch-up / ballooning	M	H	CFD C4; slow tilt rate; abort to QHOVER
R4	Tilt jam / servo stall	L	H	Stops, dual mag-angle, current trip, shaft bearings
R5	Li-ion sag under hover current	M	H	Stand + IR test; keep LiPo backup pack
R6	Pitot / IAS false transition	M	M	Fail-detect; inhibit tilt on IAS invalid
R7	EMI on compass / mag angle	M	M	Routing, shielding, GT-8
R8	Schedule slip on manufacture	H	M	COTS-first; parallel tilt bench article
R9	Loss of vehicle on first free flight	M	H	Tether, geofence, two-person crew
R10	Inconclusive power comparison	M	H	DAQ cal, repeats, locked test card

Future Improvements

V1.1

90° tilt option if 75° leaves unused axial efficiency.

V1.1

Folding props or larger disk if hover thermal is the limiter.

V2

Span / AR increase once manufacturing of the 1.3 m wing is proven.

V2

Independent motor-pod tilt (true tilt-rotor) vs paired shafts.

V2

Front/rear differential tilt as a pitch moment source.

V2

Onboard real-time CL estimator from pitot + IMU + mass.

V3

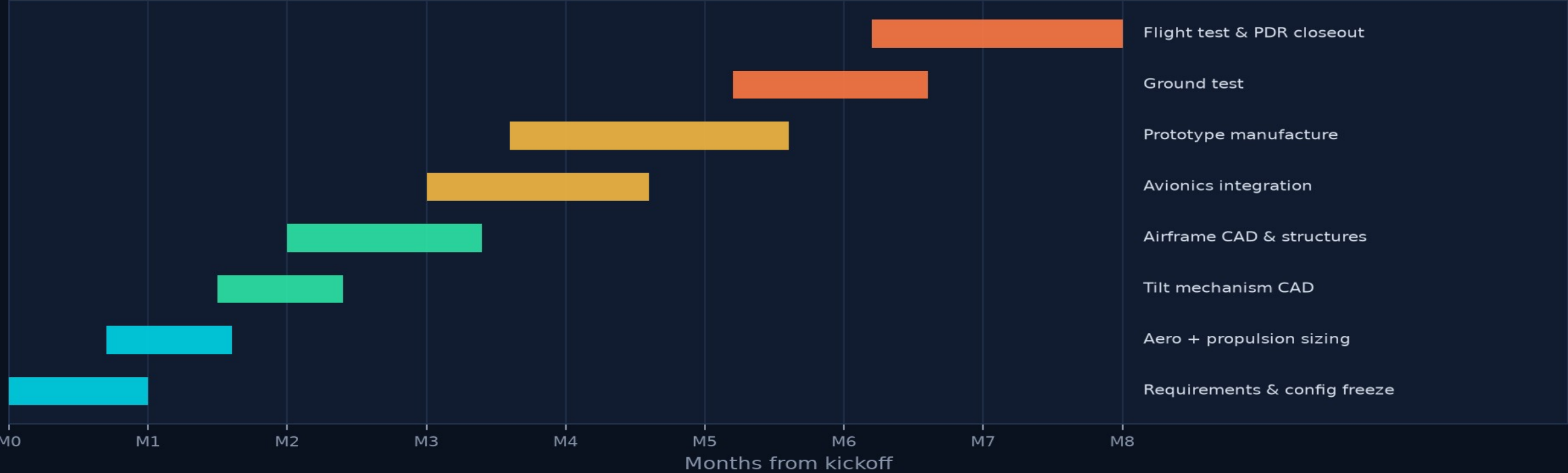
Redundant FC / dual battery if we ever leave the demo role.

V3

Mission payload (EO) replacing the 1 kg dummy — only after power is shown.

Project Timeline

8-month design-build-test programme



M2

PDR

Config, mass, thrust frozen

M5

CDR / proto

Vehicle complete, GT-10 done

M7

FT-6/7

Power comparison in hand

M8

Closeout

Report, demo, archive logs

Budget Estimate

Planning estimate · COTS-heavy · university workshop labour not costed

WBS	Group	Basis	Est. (INR)
1	Propulsion (4× MY4215, props, ESCs)	COTS	28,000
2	Battery 6S Li-ion ~20 Ah + sense	COTS	22,000
3	Pixhawk + GPS/compass + pitot	COTS	32,000
4	ESP32, harness, PDB, connectors	COTS	6,000
5	Tilt (2× STS3215, bearings, shaft)	COTS + shop	12,000
6	Mag angle sensors ×2	COTS	5,000
7	Airframe (foam/composite, prints, gear)	Shop	18,000
8	Test (stand, shunt, prints, props spares)	Shop	8,000
9	Contingency 15%	Risk	19,650
	TOTAL (planning)		≈ 1,50,650

Conclusion

Pushpak V1 is a 4.5 kg tilt-rotor VTOL demonstrator built to prove that wing-borne lift cuts electrical power versus rotor-borne flight.

01

Configuration frozen

4 rotors, paired front/rear tilt to 75°, high wing NACA 4412, Pixhawk / ArduPilot / ESP32.

02

Physics closed

Hover 1.125 kgf/motor floor, 2.25 kgf design; cruise CL 0.395 at 20 m/s on 0.32 m².

03

Experiment defined

Wing-on vs wing-off electrical power at matched mass — the only success metric that matters.

04

Path to flight

CAD → stand → tether → envelope expansion. Mass watched weekly against 3.5 kg empty.

Thank you

Questions, technical discussion, and a configuration freeze.

PROJECT TEAM

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Tilt-Rotor VTOL UAV Demonstrator

